



**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)**

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No- 11	Team No- 04	Project Title- Flight Booking Management & Analytics Portal for Airline & Airport
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1. Abstract

The Flight Booking Management & Analytics Portal for Airline & Airport is a database-driven web application designed to digitize and streamline airline and airport operations. Traditional flight booking systems often involve fragmented information, making it difficult to efficiently manage passenger details, flight schedules, bookings, baggage, check-in, and operational records.

The proposed system provides an integrated platform for managing flight booking records throughout their complete lifecycle. It allows passengers to register and manage their profiles, search and filter available flights, create and manage bookings, and handle baggage and check-in information. Role-based access control enables different users, such as passengers, airline staff, and administrators, to securely access relevant functionalities.

The system uses a centralized relational database to maintain passenger, flight, booking, baggage, check-in, and audit information while ensuring data consistency and integrity. An analytics dashboard provides meaningful insights through charts and reports, helping administrators monitor booking trends, flight operations, and passenger-related statistics. Notifications and audit logs further improve transparency and traceability.

The proposed portal aims to provide a secure, efficient, and user-friendly solution for managing airline and airport operations while demonstrating practical applications of database management concepts such as relational data modeling, normalization, transactions, queries, and role-based data access.

2.Team Details

Roll Number(s)	Name of the Student	Signature
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3. Frameworks / Platforms / Software Used

- **Frontend:** React.js
- **Backend:** Node.js, Express.js
- **Database:** MySQL
- **ORM:** Sequelize
- **Authentication & Authorization:** JWT (JSON Web Token)
- **Data Visualization:** Chart.js
- **API Development:** REST API
- **Database Management:** MySQL Workbench
- **Development Environment:** Visual Studio Code
- **Version Control:** Git and GitHub

Guide Name and Signature (remarks if any) _____